

Links

Code repository URL

https://github.com/<your-id>

Hosted app URL

https://<streamlit-share-link>

App version: v6.0-a6

Data

Choose data source

Upload CSV

Use local file (data.csv)

Will attempt to read data.csv from the working directory.

Settings

Random seed

42

Test size

0.20

Favorable outcome label

Graduate

Columns

If your column names differ, edit them below to match your file.

Target column

Target

Age column used to make AgeBin

Age at enrollment

Student Outcome Prediction — Fairness + Explainability (Assignment-6)

Extended from Assignment-4 with global and local explanations, what-if analysis, and export.

Loaded data with shape (4424, 37)

Preview data

Bias (Descriptive) by Protected Attributes

Choose attributes to summarize

Nacionality

Gender

Scholarship holder

Displaced

Debtor

Age at enrollment

AgeBin

Subgroup report for Nacionality

Nacionality	count	representation	Dropout
1	4314	0.9751	
41	38	0.0086	
26	14	0.0032	
6	13	0.0029	
22	13	0.0029	
24	5	0.0011	
100	3	0.0007	
11	3	0.0007	
103	3	0.0007	
62	2	0.0005	

Subgroup report for Gender

Gender	count	representation	Dropout
0	2868	0.6483	
1	1556	0.3517	

Subgroup report for Scholarship holder

Scholarship holder	count	representation	Dropout
0	3325	0.7516	
1	1099	0.2484	

Subgroup report for Displaced

Displaced	count	representation	Dropout
1	2426	0.5484	
0	1998	0.4516	

https://cs698y-assignment6-b7fhvqpzr9g98tv7z4tqcm.streamlit.app

1/8

Subgroup report for Debtor

Debtor	count	representation	Dropout
0	3921	0.8863	
1	503	0.1137	

Subgroup report for Age at enrollment

Age at enrollment	count	representation	Dropout
18	1036	0.2342	
19	911	0.2059	
20	599	0.1354	
21	322	0.0728	
22	174	0.0393	
24	131	0.0296	
23	108	0.0244	
26	94	0.0212	
25	93	0.0210	
27	91	0.0206	

Subgroup report for AgeBin

AgeBin	count	representation	Dropout
<=20	2551	0.5766	
21-25	828	0.1872	
>30	662	0.1496	
26-30	383	0.0866	

 Train Model

✓ Features ready!

✓ Baseline trained

Baseline Accuracy

0.7672

> Classification report (baseline)

 Fairness (Baseline)

Disparate Impact / Demographic Parity Difference (using predicted favorable rate)

Equal Opportunity Difference (TPR among true favorable)

Attribute: Gender

Gender	pred_favorable_rate	disparate_impact_vs_best	dpd_vs_best
0	0.663	1.000	0.000
1	0.451	0.680	-0.212

	TPR	EO_diff_vs_best
0	0.936	
1	0.911	

Attribute: AgeBin

AgeBin	pred_favorable_rate	disparate_impact_vs_best	dpd_vs_best
21-25	0.517	0.730	-0.191
26-30	0.375	0.530	-0.333
<=20	0.708	1.000	0.000
>30	0.370	0.523	-0.338

	TPR	EO_diff_vs_best
21-25	0.928	
26-30	0.920	
<=20	0.931	
>30	0.930	

Attribute: Scholarship holder

Scholarship holder	pred_favorable_rate	disparate_impact_vs_best	dpd_vs_best
0	0.499	0.565	-0.384
1	0.883	1.000	0.000

	TPR	EO_diff_vs_best
0	0.897	
1	0.988	

Attribute: Displaced

Displaced	pred_favorable_rate	disparate_impact_vs_best	dpd_vs_best
0	0.536	0.851	-0.094
1	0.630	1.000	0.000

	TPR	EO_diff_vs_best
0	0.950	
1	0.916	

Attribute: Debtor

Debtor	pred_favorable_rate	disparate_impact_vs_best	dpd_vs_best
0	0.629	1.000	0.000
1	0.307	0.488	-0.322

	TPR	EO_diff_vs_best
0	0.937	
1	0.833	



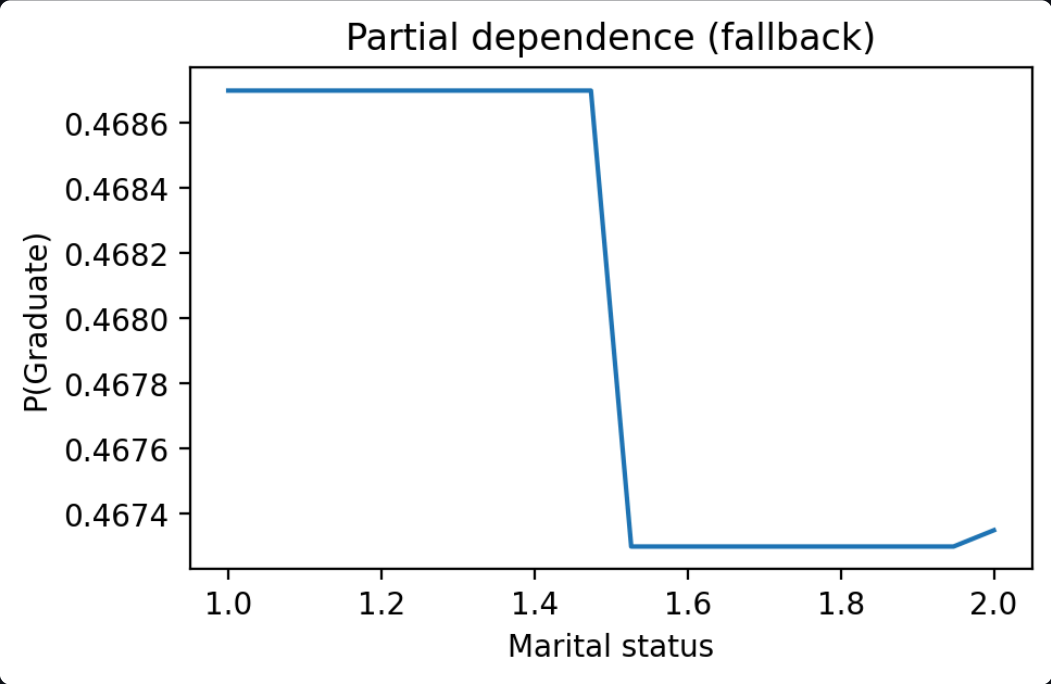
RandomForest feature importance (aggregated to base columns)

	importance
Unemployment rate	
Inflation rate	
Mother's qualification	
Father's qualification	
Curricular units 2nd sem (enrolled)	
Curricular units 1st sem (enrolled)	
Scholarship holder	
Application order	
AgeBin_<=20	
Debtor	
Gender	

✓ Computing permutation importance (global)...

Choose a numeric feature for Partial Dependence

Marital status



Local Explanations (pick a test row)

Row index from test set

2

Features (raw):

	2
Marital status	1
Application mode	1
Application order	5
Course	9070
Daytime/evening attendance	1
Previous qualification	1
Previous qualification (grade)	122.0
Nacionality	1
Mother's qualification	37
Father's qualification	37

Predicted label

Dropout

Predicted class probability (for 'Dropout'): 0.970

Install `shap` to see SHAP-based local and global explanations.

What-if analysis (counterfactual sliderboard)

Marital status
1.00

Application mode
1.00

Application order

5.00

Course

9070.00

Daytime/evening attendance

1.00

Previous qualification
1.00

Previous qualification (grade)
122.00

Nacionality
1.00

Mother's qualification

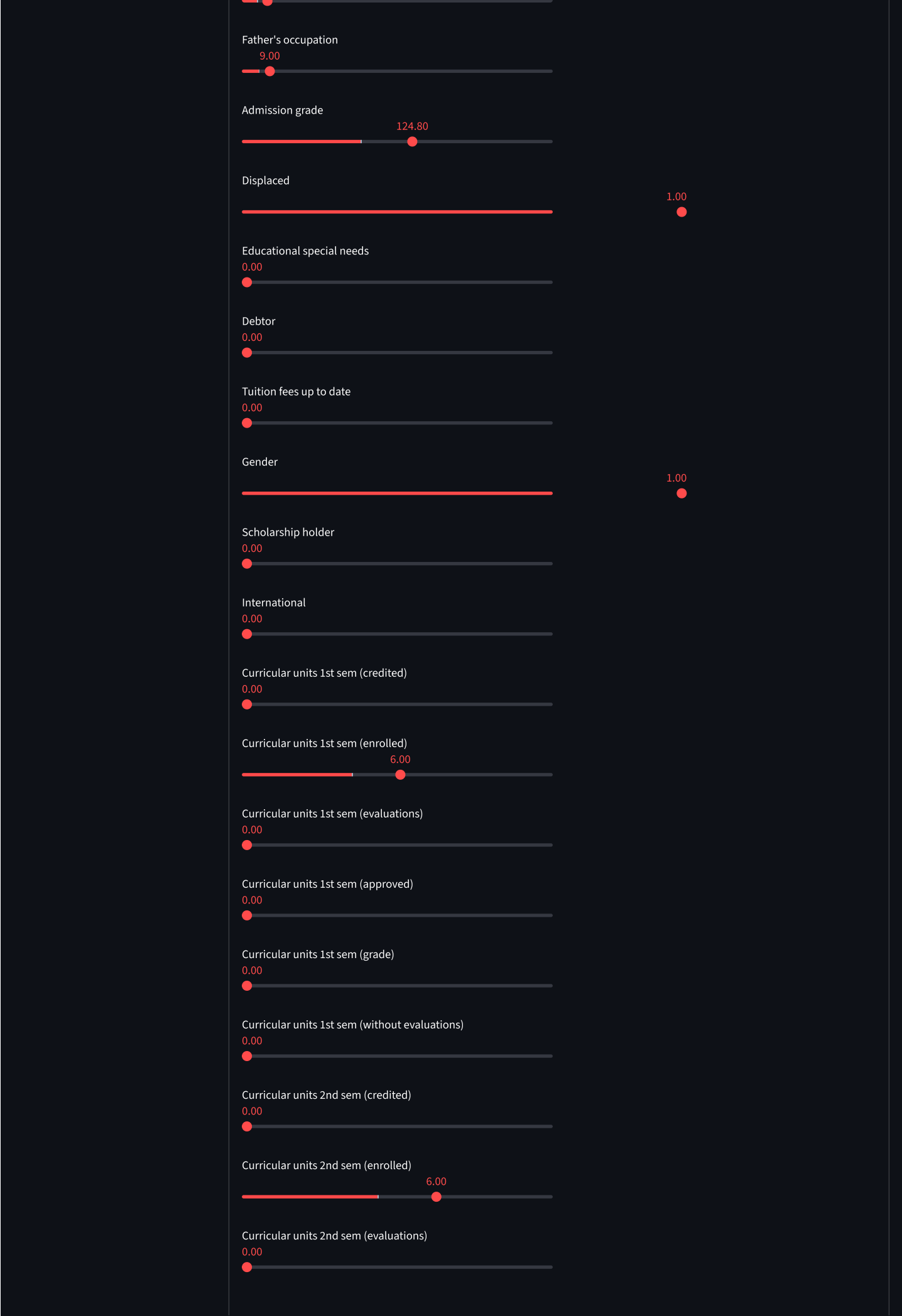
37.00

Father's qualification

37.00

Mother's occupation
9.00

Could not render selector for AgeBin:
SelectboxMixin.selectbox() got an unexpected
keyword argument 'choices'





 Bias Mitigation (Reweighting + RandomForest)

Weights groups so the intersectional distribution is more uniform within each class label. Uses groups among available protected attributes.

✓ Mitigated model trained

Mitigated Accuracy

0.7571

> Classification report (mitigated)

 Fairness (Mitigated)

Attribute: Gender

Gender	pred_favorable_rate	disparate_impact_vs_best
0	0.670	1.000
1	0.473	0.706

	TPR	EO_diff_vs_best
0	0.936	
1	0.920	

Attribute: AgeBin

AgeBin	pred_favorable_rate	disparate_impact_vs_best
21-25	0.550	0.768
26-30	0.386	0.539
<=20	0.716	1.000
>30	0.370	0.517

	TPR	EO_diff_vs_best
21-25	0.928	
26-30	0.880	
<=20	0.942	
>30	0.907	

Attribute: Scholarship holder

Scholarship holder	pred_favorable_rate	disparate_impact_vs_best
0	0.516	0.58
1	0.878	1.00

	TPR	EO_diff_vs_best
0	0.904	
1	0.981	

Attribute: Displaced

Displaced	pred_favorable_rate	disparate_impact_vs_best
0	0.539	0.828
1	0.651	1.000

	TPR	EO_diff_vs_best
0	0.939	
1	0.927	

Attribute: Debtor

Debtor	pred_favorable_rate	disparate_impact_vs_best
0	0.639	1.000
1	0.333	0.521

	TPR	EO_diff_vs_best
0	0.939	
1	0.833	



Export

Export baseline classification report JSON

Download global importances (CSV)

Tip: Use the sidebar to add your GitHub repo and hosted app links. Version label: v6.0-a6